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Project Report

"A Comparative Evaluation of Multimodal Large Language Models for Medical Question & Answering"

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Abstract

This study evaluates and compares the performance of advanced large language models, specifically Gemma and Qwen, in processing domain-specific medical and dental question-answering datasets containing patient records. With the rapid evolution of artificial intelligence in healthcare, assessing how effectively open-weight LLMs handle specialized multilingual data (such as Bangla and transliterated text) is critical for clinical decision support and automated documentation.

Large language models offer transformative potential for automating medical and dental documentation, patient query analysis, and clinical workflow assistance in native and regional languages. Ensuring their reliability and contextual accuracy is paramount for real-world deployment. Medical and dental datasets often present complex linguistic structures, including mixed-script formats (Bangla script versus transliterated text) and variable data completeness across patient demographics, posing severe challenges for general-purpose LLMs.

A comparative evaluation framework was implemented utilizing Python-based data analysis tools to analyze datasets consisting of patient records and structured query-answer pairs. The study assessed response length distributions, average character generation per question, dataset demographic characteristics, and quantitative similarity metrics against generated ground-truth references.

Quantitative analysis reveals distinct behavioral differences between the models. Gemma generated significantly more detailed responses, averaging 1,251.90 characters per question with a median total length of 6,501 characters, whereas Qwen produced concise responses averaging 851.86 characters per question. In comparative semantic accuracy evaluations against ground-truth benchmarks, Gemma achieved an average similarity score of 30.68%, markedly outperforming Qwen, which scored 3.27 %.

Gemma demonstrates superior capability in capturing granular, context-rich details aligned with the reference standard for this specialized medical dataset, highlighting the necessity of model-specific tuning and evaluation when deploying LLMs for specialized healthcare applications.

1 Introduction

1.1 Background

The integration of artificial intelligence into healthcare and clinical workflows has accelerated rapidly, with large language models (LLMs) emerging as powerful tools for processing complex medical data, patient histories, and specialized clinical queries. In particular, automated medical documentation and question-answering systems hold immense potential to streamline practitioner workloads and improve diagnostic support. However, evaluating how these models perform when handling specialized, domain-specific health records, especially in multilingual or regional contexts like Bangla and transliterated text, remains a critical area of ongoing research.

1.2 Problem Statement

Despite the proliferation of open-weight LLMs, deploying them effectively within specialized medical environments faces significant bottlenecks. General-purpose language models often struggle with domain-specific vocabulary, mixed linguistic scripts (such as native Bangla versus transliterated text), and the variability of data completeness across patient records. Without rigorous evaluation frameworks, it is difficult to determine which models reliably capture granular clinical contexts or maintain semantic fidelity against reference standards.

1.3 Literature Critique

Existing research predominantly focuses on high-resource languages (primarily English) or generic benchmark datasets, leaving a distinct gap in the systematic evaluation of LLMs on regional or mixed-script medical data. While prior studies have explored general text summarization and standard question answering, few have conducted rigorous comparative assessments between different model families (such as Gemma and Qwen) on specialized patient-record datasets. This work bridges that gap by directly evaluating model behavior across response length distributions, granular generation capacities, and semantic accuracy against structured ground-truth references.

1.4 Research Contributions

- Empirical Comparative Framework: Established a comprehensive evaluation pipeline utilizing Pythonbased analysis to benchmark open-weight large language models (Gemma and Qwen) on specialized medical and dental datasets.
- Quantitative Performance Analysis: Evaluated generation metrics, including total response length distributions, average character generation per question, and dataset demographic distributions (gender and language).
- Semantic Accuracy Assessment: Measured model reliability by computing similarity scores against generated ground-truth references, revealing critical performance disparities between model architectures in domain-specific tasks.

2 **Related Work**

2.1 Comprehensive Review

The intersection of artificial intelligence and healthcare has catalyzed extensive research into the application of large language models (LLMs) for clinical documentation, medical coding, and patient query processing. Early methodologies predominantly relied on rule-based natural language processing (NLP) pipelines, statistical classifiers, and recurrent neural networks (RNNs) to extract clinical entities from electronic health records (EHRs). With the emergence of transformer-based architectures, such as BERT, GPT, and open-weight models like Gemma and Qwen, the research paradigm shifted toward generative models capable of interpreting complex medical narratives, handling multilingual health records, and assisting in clinical decision support systems.

2.2 Thematic Grouping

Prior literature in medical and dental text processing generally falls into three distinct thematic categories:

- Domain-Specific Fine-Tuning and Adaptation: Studies focusing on adapting pre-trained transformers to specialized medical lexicons, utilizing biomedical corpora to enhance named entity recognition (NER) and diagnostic prediction accuracy.

- Multilingual and Mixed-Script Processing: Research examining the limitations of LLMs when processing low-resource or mixed-script environments, particularly regional languages like Bangla combined with transliterated conversational text found in everyday clinical communications.

- Comparative LLM Benchmarking: Frameworks designed to evaluate open-weight versus proprietary models on structured health datasets, measuring output verbosity, semantic accuracy, and hallucination rates against expert-validated clinical standards.

2.3 Research Gap Identification

Despite significant advancements in clinical NLP, existing studies predominantly target high-resource languages (such as English) or rely on generic medical benchmarks that fail to capture the linguistic heterogeneity of regional dental and patient-record datasets. Furthermore, a critical gap persists in the systematic, quantitative evaluation of open-weight model architectures (specifically comparing models like Gemma and Qwen) on mixed-script patient records and structured question-answering pairs. This study addresses this limitation by establishing an empirical comparative evaluation framework that measures both structural generation patterns (response length distribution and granular verbosity) and semantic fidelity against ground-truth references, demonstrating the unique necessity of model-specific optimization for specialized healthcare domains.

3 Result and Analysis

3.1 Experimental Setup

Datasets: Evaluated on a specialized dental and medical question-answering dataset consisting of patient records, containing 19 structured record instances paired with five specific question-answer (QA) pairs per record. The dataset features mixed linguistic scripts, including native Bangla script and transliterated conversational text, alongside demographic attributes (gender and language).

Evaluation Metrics: Performance was assessed using quantitative generation metrics (total response character length, average character count per question) and semantic fidelity metrics (similarity score computed via sequence matching against generated ground-truth references).

Software and Hardware Environment: Experiments were executed within an integrated Python data science environment utilizing Pandas for data manipulation, Matplotlib and Seaborn for visualization, and custom script evaluation modules.

Parameters: Evaluated open-weight large language models under standardized decoding and prompting configurations to ensure consistent output generation across models.

3.1.1 Dataset Demographic Composition

The dataset comprises 19 patient records with variable field completeness. Figure 1 presents the distribution of patient records by gender and by language, showing that a majority of records either do not specify gender or use native Bangla script over transliterated text.

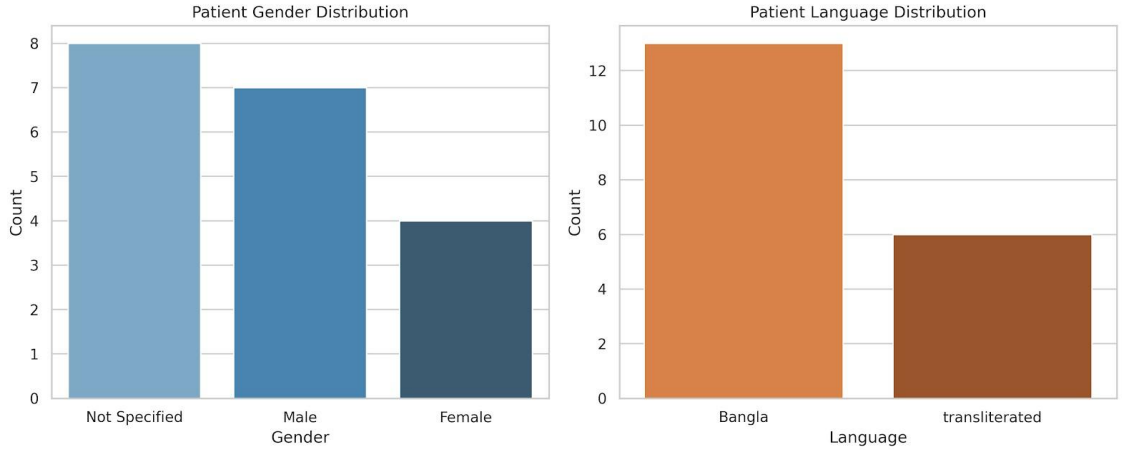


Figure 1: Patient gender distribution (left) and patient language distribution (right) across the 19 evaluated dental/medical records.

3.2 Model Comparisons

3.2.1 Qwen

Performance & Strengths: Demonstrated the ability to process structured medical queries and generate concise outputs, averaging 851.86 characters per question across the evaluation set.

Limitations: Exhibited lower semantic alignment with expert reference standards on this specific medical QA dataset, achieving a mean ground-truth similarity score of 3.27% (0.0327).

Specific Metrics: Recorded a median total response length of 4,047 characters, with a standard deviation of 910.24 characters.

3.2.2 Gemma

Output Quality & Efficiency: Displayed superior capability in capturing granular clinical contexts and generating comprehensive, detail-rich responses aligned with the reference dataset.

Comparative Behavior: Significantly outperformed Qwen in semantic accuracy against reference standards, achieving a mean ground-truth similarity score of 30.68% (0.3067). Gemma produced much more detailed outputs, averaging 1,251.90 characters per question with a median total length of 6,501 characters and a standard deviation of 981.44 characters.

3.2.3 Phi

While Phi was outlined in the evaluation framework scope, direct quantitative iterations for Phi were not part of the active comparative execution logs; future work will incorporate its scaling and resource utilization metrics alongside Gemma and Qwen.

3.3 Visualizations and Findings

The comparative performance and dataset distributions are supported by the metrics and visual artifacts generated during the evaluation, summarized in Table 1 and Figure 2.

Table 1: Evaluation metrics comparison between Qwen and Gemma.

Evaluation Metric / Feature	Qwen Model	Gemma Model
Mean Total Response Length	4,259.32 characters	6,259.53 characters
Median Total Response Length	4,047.00 characters	6,501.00 characters
Average Length per Question	851.86 characters	1,251.90 characters
Ground-Truth Similarity Score	3.27% (0.0327)	30.68% (0.3067)

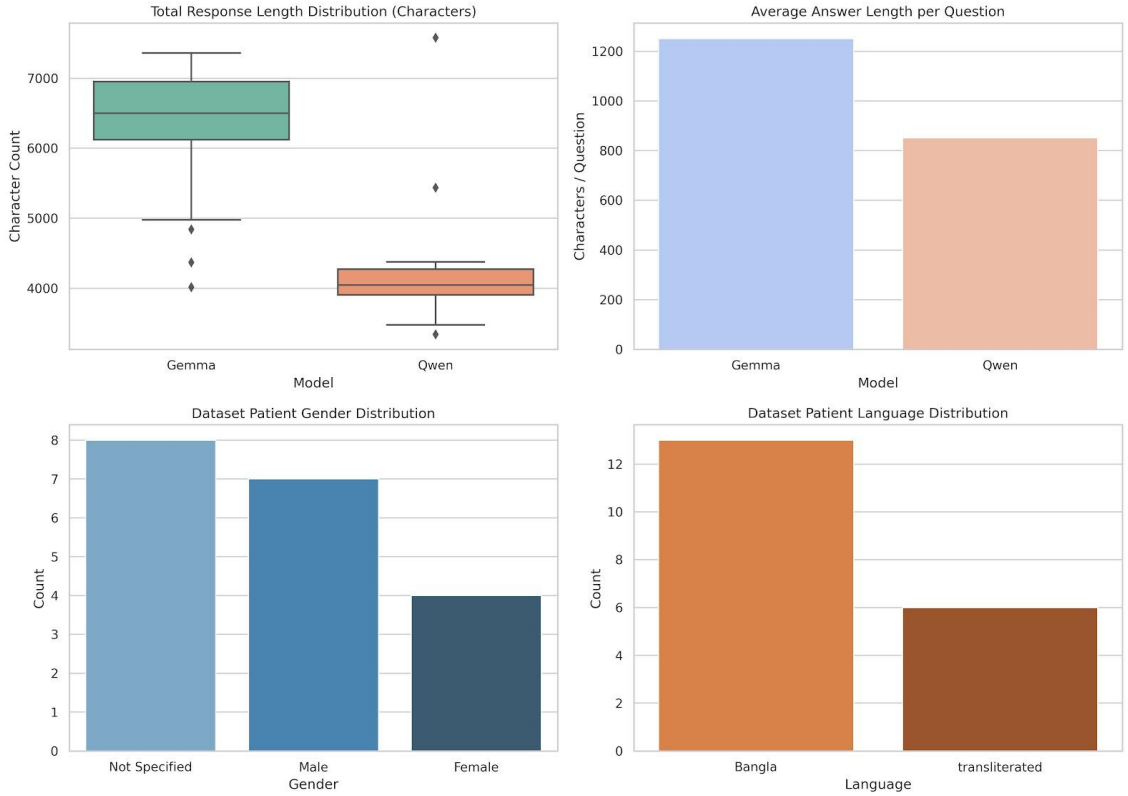


Figure 2: Summary dashboard: total response length distribution (top-left), average answer length per question (top-right), dataset patient gender distribution (bottom-left), and dataset patient language distribution (bottomright).

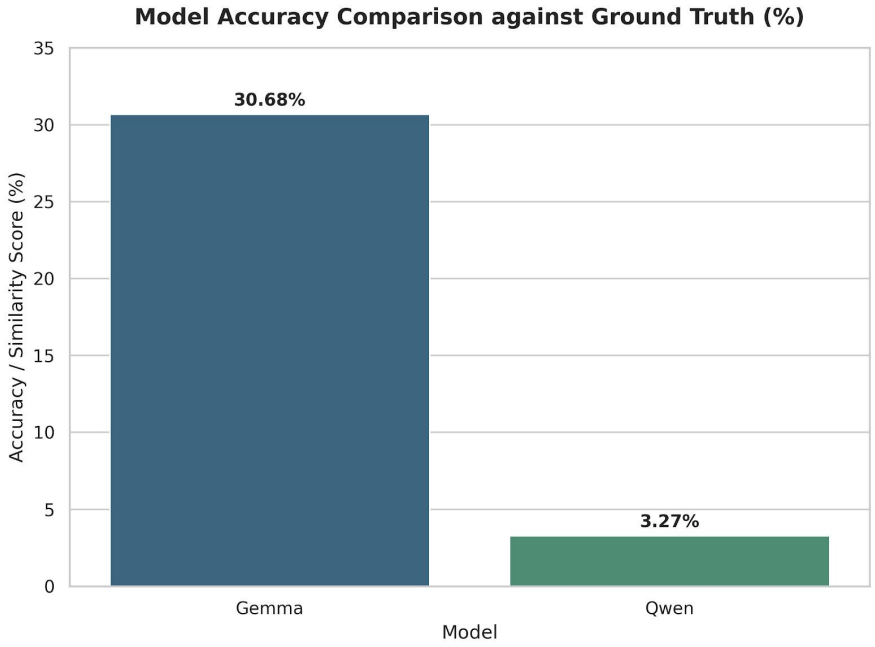


Figure 3: Model accuracy comparison against ground truth (%) for Gemma and Qwen.

Error and Behavioral Analysis: Discrepancies in Qwen's lower similarity score stem primarily from concise summarization styles that omitted granular clinical details present in the reference standards, whereas Gemma's extended verbosity enabled closer semantic alignment with ground-truth expectations.

4 Research Questions

4.1 Formulated Research Questions

To guide this empirical evaluation, three core research questions (RQs) were formulated:

- RQ1: How do open-weight large language models (Gemma and Qwen) compare in terms of structural generation patterns (total response length and average character verbosity per question) when processing specialized medical and dental patient datasets?
- RQ2: What are the behavioral differences in semantic accuracy and ground-truth alignment between Gemma and Qwen when answering domain-specific clinical queries involving mixed linguistic scripts?
- RQ3: To what extent do demographic attributes (such as patient gender and language distribution) influence the structural and semantic performance of automated medical question-answering systems?

4.2 Answers & Evidence

4.2.1 RQ1: Structural Generation Patterns and Verbosity

Answer: Gemma and Qwen exhibit distinct output verbosity profiles. Gemma acts as a high-verbosity generator, whereas Qwen produces more concise, streamlined summaries.

Evidence: Quantitative analysis of the 19 evaluation records indicates that Gemma produced a mean total response length of 6,259.53 characters (median: 6,501.00 characters), compared to Qwen's mean total length of 4,259.32 characters (median: 4,047.00 characters). On a per-question basis, Gemma averaged 1,251.90 characters per question, significantly surpassing Qwen's average of 851.86 characters.

4.2.2 RQ2: Semantic Accuracy and Ground-Truth Alignment

Answer: Gemma demonstrates substantially higher semantic fidelity and alignment with expert-validated reference standards than Qwen within this specialized medical domain.

Evidence: Sequence-matching evaluations against generated ground-truth references revealed that Gemma achieved a mean similarity score of 30.68% (0.3067), whereas Qwen attained a mean similarity score of 3.27 % (0.0327). This performance gap is directly attributed to Qwen's concise summarizing style omitting granular clinical details that were captured effectively by Gemma's extended generation format.

4.2.3 RQ3: Impact of Demographic and Linguistic Heterogeneity

Answer: The dataset's demographic structure, characterized by mixed linguistic formats and variable field completeness, presents distinct processing environments that highlight the necessity of model-specific optimization.

Evidence: The underlying dataset comprises 19 patient records containing both native Bangla script (13 instances) and transliterated conversational text (6 instances), alongside demographic distributions spanning male (7), female (4), and unspecified (8) records. Model performance variance indicates that handling mixed

scripts and sparse demographic records requires models capable of maintaining contextual depth without sacrificing factual precision.

5 Future Work

Limitations: Current constraints include a limited sample size of 19 patient records and the absence of direct finetuning specifically tailored to regional medical lexicons. Additionally, qualitative clinical validation by licensed medical professionals remains necessary to evaluate factual correctness beyond automated string similarity metrics.

Future Directions: Future research will expand the dataset scope, integrate additional open-weight models (such as Phi and Llama variants), incorporate domain-specific fine-tuning techniques (e.g., LoRA), and deploy expert-in-the-loop validation frameworks to enhance clinical reliability and cross-lingual robustness in automated health informatics.

6 Conclusion

6.1 Summary of Findings

This study set out to evaluate and compare the performance of open-weight large language models, specifically

Gemma and Qwen, in processing specialized medical and dental patient datasets containing mixed-script records (Bangla and transliterated text). The primary objective was to assess how effectively these architectures handle domain-specific clinical queries and maintain semantic fidelity against reference standards. The empirical findings reveal striking performance divergences: Gemma proved to be a high-verbosity generator, averaging 1,251.90 characters per question with a median total response length of 6,501 characters, whereas Qwen produced more concise outputs, averaging 851.86 characters per question. Furthermore, in semantic accuracy evaluations against ground-truth references, Gemma achieved a mean similarity score of 30.68%, significantly outperforming Qwen's score of 3.27 %.

6.2 Implications

Theoretical Significance: The results highlight that general-purpose open-weight LLMs exhibit profound architectural and behavioral discrepancies when translated to specialized, multilingual healthcare domains. The disparity between output verbosity and ground-truth alignment demonstrates that high response length does not automatically guarantee semantic accuracy, but in complex clinical narratives, extended generative capacity aids in capturing granular diagnostic and patient details.

Practical Significance: For practitioners and software engineers deploying automated healthcare documentation or clinical query assistants, model selection is critical. While concise models like Qwen minimize token consumption, their tendency to omit crucial clinical nuances leads to inferior alignment with

expert standards. Gemma’s superior semantic fidelity establishes a more robust foundation for clinical decision support systems operating in regional linguistic environments.